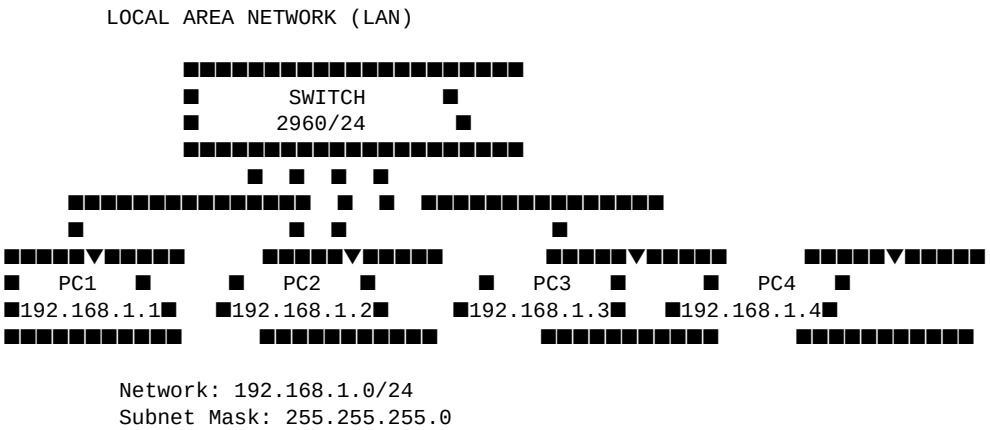
EXPERIMENT 4

LAN Configuration

1. Aim

To set up a Local Area Network (LAN) by linking several computers to a switch, assigning an IP address to each device, and checking that the connected devices can communicate.

2. LAN Diagram



3. IP Configuration

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.1.1	255.255.255.0	192.168.1.254
PC2	192.168.1.2	255.255.255.0	192.168.1.254
PC3	192.168.1.3	255.255.255.0	192.168.1.254
PC4	192.168.1.4	255.255.255.0	192.168.1.254
Switch	192.168.1.254	255.255.255.0	—

Network Address: 192.168.1.0
Broadcast Address: 192.168.1.255
Subnet Mask: 255.255.255.0

4. Procedure

1. Connect each computer to the switch with an Ethernet cable.
2. Set a different IP address for each computer.
3. Apply the same subnet mask to all computers.
4. Enter the default gateway when it is needed.
5. Confirm that the switch ports are active.
6. Run the ping command to check connectivity among the computers.
7. Confirm that data packets are being sent successfully between devices.

Example ping tests:

```
PC1 → ping 192.168.1.2
PC1 → ping 192.168.1.3
PC1 → ping 192.168.1.4
```

If the computers return successful replies, the LAN configuration is working as expected.

5. Outcome

The Local Area Network was established by connecting multiple computers to a switch. A unique IP address was configured for each device, and communication between the computers was checked successfully with the ping command.

6. Challenges Faced

• Providing a unique IP address to every connected device.

• Making sure all devices were configured within the same network.

• Verifying that the subnet mask was entered correctly.

• Finding and correcting errors in IP address settings.

• Investigating connectivity problems when a device failed to reply to ping.

• Checking Ethernet cables and switch-port connections.

• Preventing duplicate IP addresses on the network.

7. Result

Therefore, the Local Area Network was configured successfully. Communication among the connected devices was established and verified.